

peak_ewma_p2c load test (#94)

cluster-2026-10-05-fixes · commit 1eb5caf · Darwin 25.3.0 arm64 · 18 CPUs

- CPU per RPC at the centre point: peak_ewma_p2c 30.50 μ s vs round_robin 30.81 μ s (-1.0%); LB cost above the no-LB control: 7.27 μ s vs 7.58 μ s.
- Allocation: 249 B/RPC more than round_robin (+0.8%).
- Brownout p99 during the fault: 21.5 ms vs round_robin 450.0 ms (ratio 0.05 $\times$).
- Steady heterogeneous p99: 26.9 ms vs round_robin 93.8 ms (ratio 0.29 $\times$).
- Crash-loop error rate during the fault: 0.05% vs round_robin 4.87%.
- Black hole error rate during the fault: 0.09% vs round_robin 4.81%.

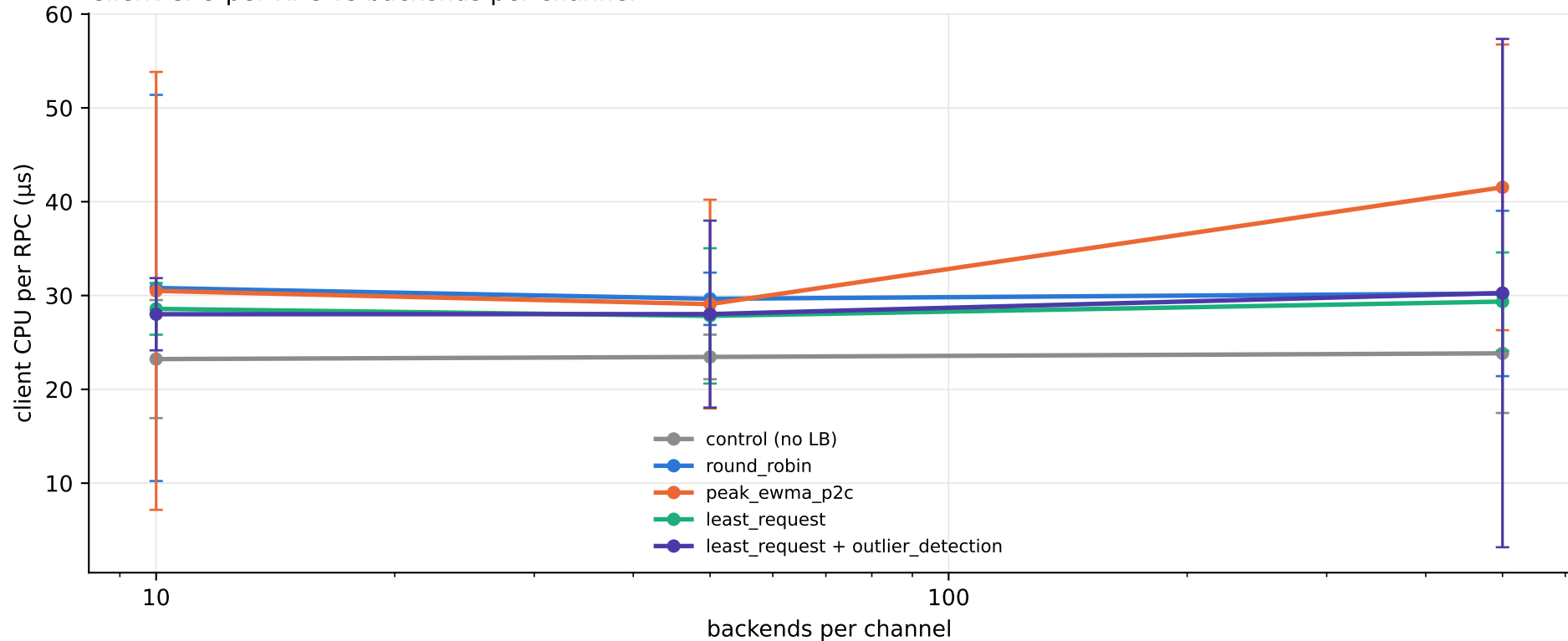
Decision table (worst-case p99 / error rate per scenario)

scenario	p99 peak (ms)	p99 least_request (ms)	p99 lr+od (ms)	p99 round_robin (ms)	errors peak	errors least_request	errors lr+od	errors round_robin	verdict (peak vs least_request)
s01_steady_het	26.9	76.1	76.0	93.8	0.03%	0.29%	0.07%	0.27%	peak: p99 65% lower than lr+od
s02_steady_hom	21.3	21.4	21.3	21.3	0.00%	0.00%	0.00%	0.00%	tie
s03_brownout	21.5	22.6	21.5	450.0	0.03%	0.15%	0.10%	1.50%	least_request: recovers faster (4 s vs 10 s)
s04_crashloop	21.5	21.4	21.3	21.3	0.05%	7.59%	1.87%	4.87%	peak: fewer errors (0.05% vs 1.87%);
s05_blackhole	21.5	21.5	21.4	21.3	0.09%	0.47%	0.12%	4.81%	tie
s10_node_cpu_cd	23.6	332.8	314.1	487.9	0.14%	0.52%	0.47%	2.36%	peak: p99 92% lower than lr+od; peak: p99 60% lower than least_request
s11_client_restart	30.3	76.5	77.2	97.8	0.04%	0.30%	0.09%	0.27%	peak: p99 60% lower than least_request

Go / no-go

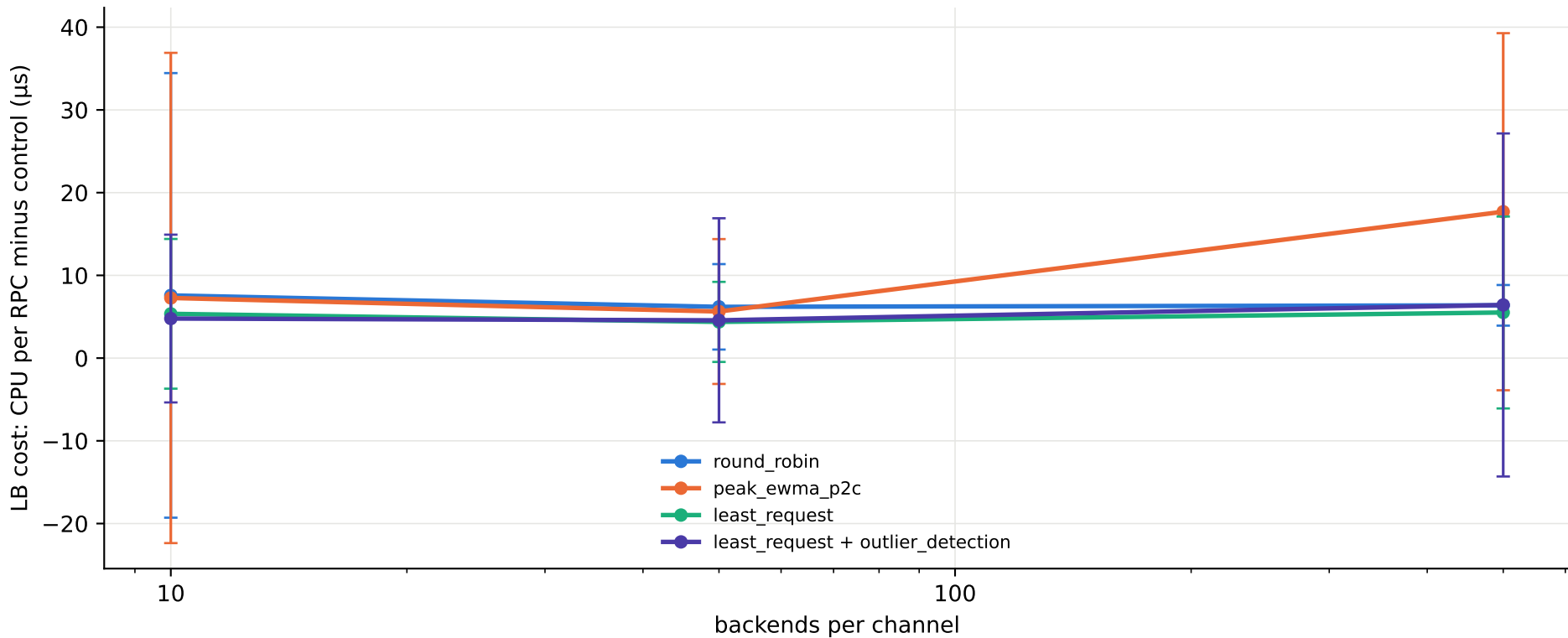
area	criterion	measured	result	note
CPU overhead	client CPU/RPC $\leq +25\%$ vs round_robin	30.50 vs 30.81 $\mu\text{s}/\text{RPC}$ (-1.0%)	PASS	whole-process CPU incl. Netty + harness; LB control
CPU overhead	client CPU/RPC $\leq +25\%$ vs round_robin	29.09 vs 29.65 $\mu\text{s}/\text{RPC}$ (-1.9%)	PASS	whole-process CPU incl. Netty + harness; LB control
Allocation	bytes/RPC $\leq +10\%$ vs round_robin	32,507 vs 32,258 B/RPC (+0.8%)	PASS	harness floor $\approx$ control's B/RPC dilutes the %; threshold
Memory	LB heap plateaus under 1,000 MB	no data	N/A	
Pick time	p99 pick < 5 μs at 500 backlog	9,336 ns (sampled 1/1000)	FAIL	wall time of pickSubchannel incl. the timer itself
Steady heterogeneous	p99 better than round_robin; errors 0	p99 26.9 vs 93.8 ms; errors 0	PASS	
Brownout	detected within 10 s; p99 during fault	detect 3.0 s; p99 during 21.5 s	PASS	p99 is of successful calls; errors during the fault
Crash-loop	client error rate $\leq 20\%$ of round_robin	0.054% vs 4.872% (0.01 $\times$)	PASS	
Black hole	client error rate $\leq 20\%$ of round_robin	0.092% vs 4.813% (0.02 $\times$)	PASS	
Homogeneous steady	0 false ejections/hour; every client	0 ejections in 6.0 client-minutes	PASS	
Herd (16 clients)	max healthy-backend share $\leq 10\%$	no data	N/A	
vs least_request + outlier_detection	document where each wins	s01: p99 peak (26.9/76.0 ms)	INFO	

Client CPU per RPC vs backends per channel



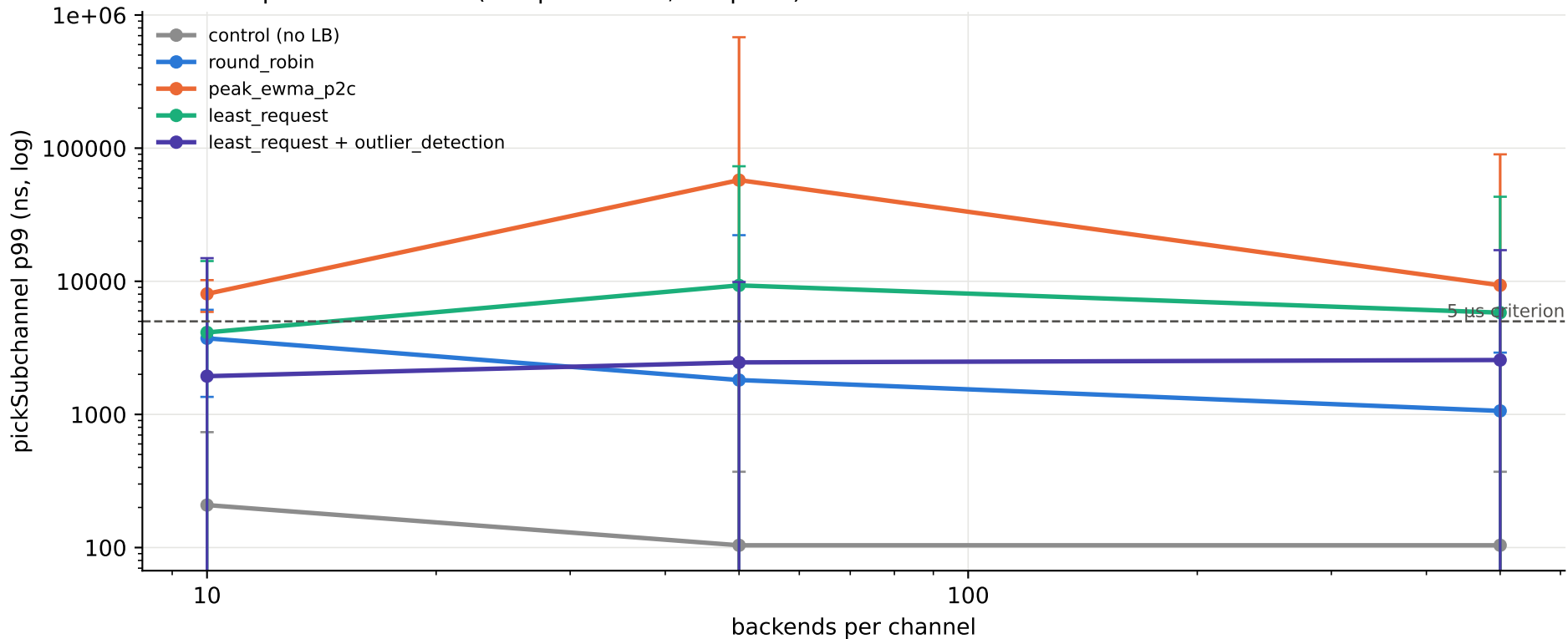
At 500 backends per channel, peak_ewma_p2c is 41.53 vs round_robin 30.22 (+37%).

LB cost (policy - control) vs backends



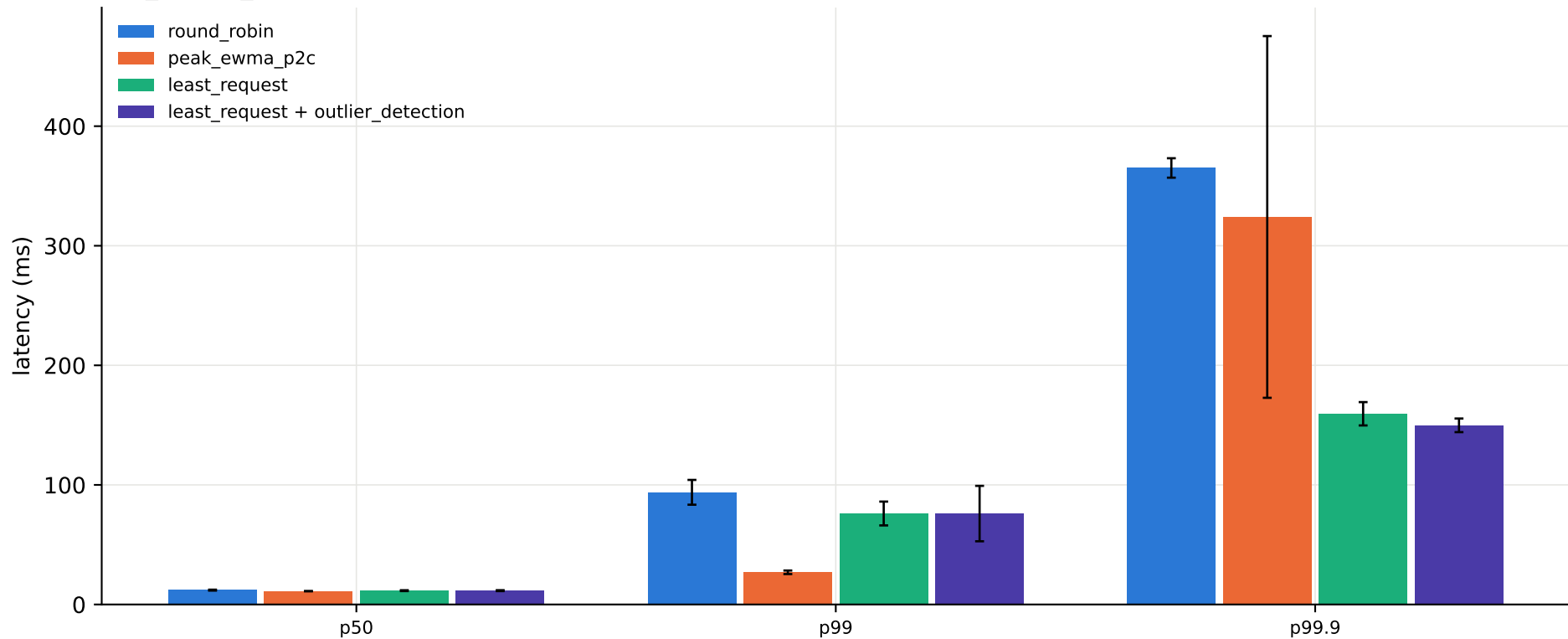
At 500 backends per channel, peak_ewma_p2c is 17.70 vs round_robin 6.38 (+177%).

Pick time p99 vs backends (sampled 1 in 1,000 picks)



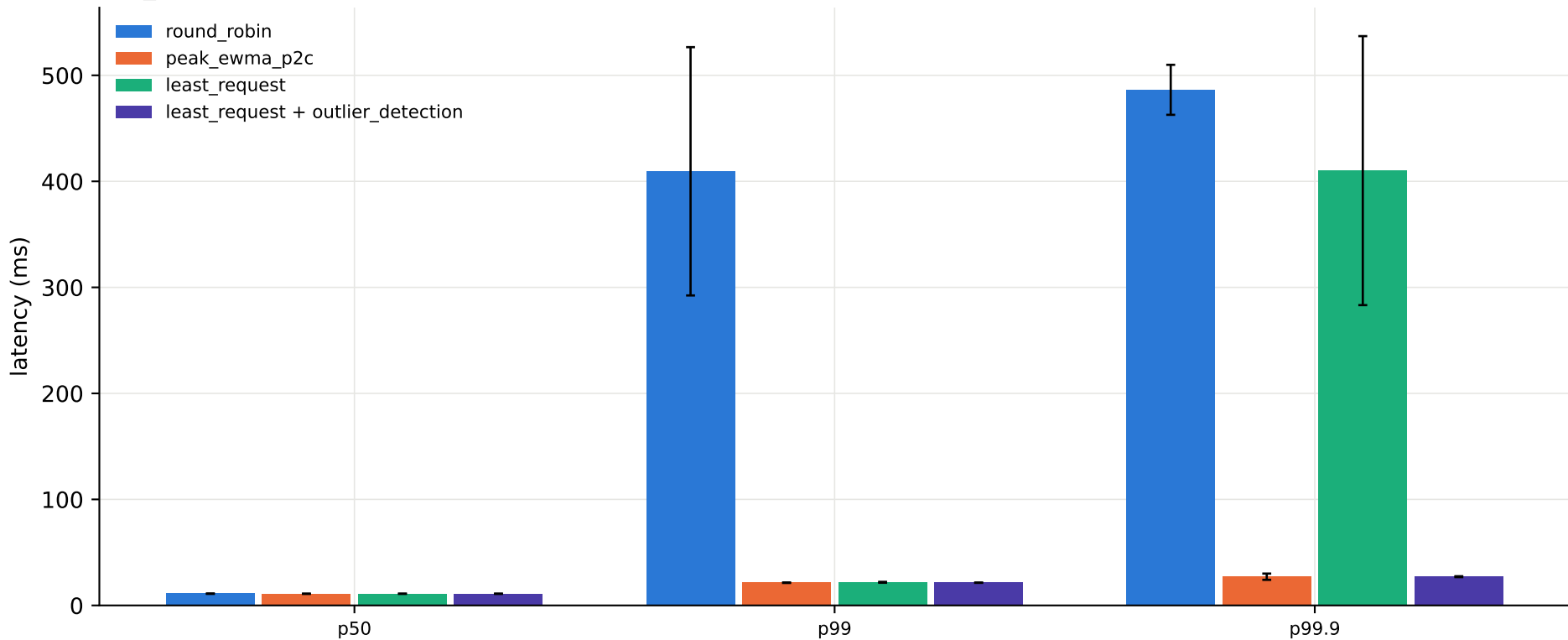
At 500 backends per channel, peak_ewma_p2c is 9336.00 vs round_robin 1062.50 (+779%).

s01_steady_het: client latency percentiles (whole window)



p99: peak_ewma_p2c 26.9 ms vs round_robin 93.8 ms (-71%); lowest p99: peak_ewma_p2c.

s03_brownout: client latency percentiles (whole window)



p99: peak_ewma_p2c 21.4 ms vs round_robin 409.5 ms (-95%); lowest p99: peak_ewma_p2c.

s03_brownout: p99 over time (U10_get, 1 s windows, repeat 0)

